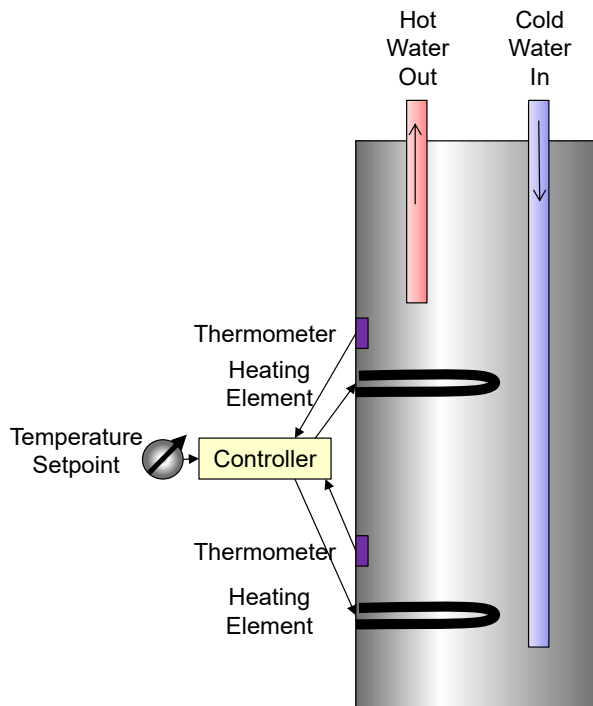

CSE211s Introduction to Embedded Systems

Spring 2023

Sheet 2

Introduction to Embedded Systems

1. Consider a controller for an electric water heater (as shown below) as an embedded computer system.



- a. What are the input devices?
 - b. What are the outputs?
 - c. Name at least two safety features to include, and specify what hardware and/or software is needed for each.
 - d. Describe a useful feature which could you add in software without requiring additional hardware.
 - e. Describe two useful energy-saving features which could you add in software if the controller could keep tracking the time of day.
 - f. Describe a useful feature which could you add in software if the water heater included an internet connection.
2. You have been asked to add an “automatic power-off” controller to a flashlight. The flashlight should turn off automatically after 3 minutes if it is not being used any more. The controller must be small, inexpensive, and use little power.
 - a. Can you think of a way to do this with electronics but without an embedded computer?
 - b. Can you think of a way to do this with a mechanical approach?
 - c. Now, really thinking outside the box, can you think of a way to do this with a pneumatic approach (using air)?

3. Consider embedding a computer in a flashlight. List three benefits or new useful features that are now possible. How much more would you be willing to pay for each feature?
4. A modern automobile has dozens of embedded computers. However, the auto industry is extremely cost-sensitive, so there must be a compelling reason to add computers. Give five examples of features enabled by embedded computers, and explain what major benefit(s) they provide.
5. Look around and find five devices that are likely to have embedded computer control systems. Answer these questions for each device:
 - a. What does the device do?
 - b. What are the device's inputs?
 - c. What are the device's outputs?
 - d. How does the embedded computer improve the device? Does it provide more features? Does it improve performance?
 - e. What are the biggest constraints on the device (size, cost, power, etc.)?
 - f. Would it be possible to build the device using a different kind of controller (e.g. mechanical)? How would that affect the device's features, performance, cost, size, and weight?